RACE Regular Analysis Through Continuous Exercise

Maxima and Minima : RACE-01

Single Correct Question

1. $f(x) = \begin{cases} \tan^{-1} x, & |x| < \frac{\pi}{4} \\ \frac{\pi}{2} - |x|, & |x| \geq \frac{\pi}{4} \end{cases}$, then

(A) $f(x)$ has no point of local maxima

(B) $f(x)$ has only one point of local maxima

(C) $f(x)$ has exactly two points of local maxima

(D) $f(x)$ has exactly two points of local minima

MMM133

2. Let $f(x) = \begin{cases} |x-2| + a^2 - 9a - 9 & \text{if} \quad x < 2 \\ 2x - 3 & \text{if} \quad x \geq 2 \end{cases}$

Then find the value of 'a' for which $f(x)$ has local minimum at $x = 2$

(A) $(-\infty, 0] \cup [1, \infty)$ (B) $(-\infty, -1] \cup [10, \infty)$

(C) $[3, \infty)$ (D) $[-1, 10]$

MMM134

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3. If $f(x) = \begin{cases} 3 + |x-k|, & x \leq k \\ a^2 - 2 + \frac{\sin(x-k)}{x-k}, & x > k \end{cases}$ has minimum at $x = k$, then

(A) $a \in R$ (B) $|a| < 2$ (C) $|a| > 2$ (D) $1 < |a| < 2$

MMM135

4. The function $x\sqrt{1-x^2}, (x > 0)$ has

(A) A local maxima

(B) A local minima

(C) Neither a local maxima nor a local minima

(D) None of these

MMM136

5. The point for the curve $y = xe^x$

(A) $x = -1$ is minimum (B) $x = 0$ is minimum

(C) $x = -1$ is maximum (D) $x = 0$ is maximum

MMM137

6. The minimum value of the function defined by $f(x) = max\,(x, x+1, 2-x)$ is

(A) 0 (B) 1/2 (C) 1 (D) 3/2

MMM138

7. The function $sin\,x\,(1 + cos\,x)$ at $x = \frac{\pi}{3}$, is

(A) Maximum (B) Minimum

(C) Neither maximum nor minimum (D) None of these

MMM139

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8. If for a function $f(x), f'(a) = 0, f'''(a) \neq 0$, then at $x = a, f(x)$ is

(A) Minimum (B) Maximum

(C) Not an extreme point (D) Extreme point

MMM140

9. For what value of $x, x^2\,\ell n(1/x)$ is maximum-

(A) $e^{-\frac{1}{2}}$ (B) $e^{\frac{1}{2}}$ (C) e (D) e^{-1}

MMM141

10. The function $f(x) = a\sin x + \frac{1}{3}\sin 3x$ has a maximum at $x = \pi/3$, then a equals

(A) – 2 (B) 2 (C) – 1 (D) 1

MMM142

11. Function $f(x) = e^x + e^{-x}$ has -

(A) One minimum point (B) One maximum point

(C) Many extreme points (D) No extreme point

MMM143

12. If $f(x) = 1 + 2x^2 + 4x^4 + 6x^6 + \ldots\ldots + 100x^{100}$ is a polynomial in a real variable x, then $f(x)$ has:

(A) neither a maximum nor a minimum

(B) only one maximum

(C) only one minimum

(D) one maximum and one minimum

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MMM144

13. The function $f(x) = \sum_{k=1}^{5}(x-k)^2$ assumes minimum value for x given by-

(A) 5 (B) 3 (C) 5/2 (D) 2

MMM145

Multiple Correct Question

14. Let $f(x) = (x^2-1)^n\,(x^2+x+1).\,f(x)$ has local extremum at $x = 1$ if

(A) $n = 2$ (B) $n = 3$ (C) $n = 4$ (D) $n = 6$

MMM146

15. If $f(x) = \frac{x}{1+x\tan x}, x \in \left(0, \frac{\pi}{2}\right)$, then

(A) $f(x)$ has exactly one point of minima

(B) $f(x)$ has exactly one point of maxima

(C) $f(x)$ is increasing in $\left(0, \frac{\pi}{2}\right)$

(D) maxima occurs at x_0 where $x_0 = \cos x_0$

MMM147

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16. A function $f(x) = \frac{x^2-3x+2}{x^2+2x-3}$ is -

(A) 1 is not in its domain

(B) minimum at $x = -3$ and maximum at $x = 1$

(C) no point of maxima and minima

(D) increasing in its domain

MMM148

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	C	B	C	A	A	D	A	C	A	B
Question	11	12	13	14	15	16				
Answer	A	C	B	A,C,D	B,D	A,C				

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RACE
Regular Analysis Through Continuous Exercise

Maxima and Minima : RACE-02

Single Correct Question

1. The adjacent sides of a rectangle with given perimeter as 100 *cm* and enclosing maximum area are
(A) 10 cm and 40 cm (B) 20 cm and 30 cm
(C) 25 cm and 25 cm (D) 15 cm and 35 cm

MMM149

2. The area of a rectangle will be maximum for the given perimeter, when rectangle is a
(A) Parallelogram (B) Trapezium (C) Square (D) None of these

MMM150

3. Of the given perimeter, the triangle having maximum area is
(A) Isosceles triangle (B) Right angled triangle
(C) Equilateral (D) None of these

MMM151

4. If from a wire of length 36 *metre* a rectangle of greatest area is made, then its two adjacent sides in metre are
(A) 6, 12 (B) 9, 9 (C) 10, 8 (D) 13, 5

MMM152

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5. The perimeter of a sector is p. The area of the sector is maximum when its radius is
(A) $\sqrt{p}$ (B) $\frac{1}{\sqrt{p}}$ (C) $\frac{p}{2}$ (D) $\frac{p}{4}$

MMM153

6. If $ab = 2a + 3b, a > 0, b > 0$ then the minimum value of ab is
(A) 12 (B) 24 (C) $\frac{1}{4}$ (D) None of these

MMM154

7. If PQ and PR are the two sides of a triangle, then the angle between them which gives maximum area of the triangle is
(A) π (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{2}$

MMM155

8. 20 is divided into two parts so that product of cube of one quantity and square of the other quantity is maximum. The parts are-
(A) 10, 10 (B) 16, 4 (C) 6, 14 (D) 12, 8

MMM156

9. The ratio between the height of a right circular cone of maximum volume inscribed in a given sphere and the diameter of the sphere is-
(A) 2 :3 (B) 3 :4 (C) 1 :3 (D) 1 : 4

MMM157

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10. A triangle with maximum area inscribed in a circle is-

(A) Right angled (B) Isosceles

(C) equilateral (D) Isosceles right angled

MMM158

Multiple Correct Question

11. Let $f(x)=\frac{40}{3x^4+8x^3-18x^2+60}$. Which of the following statement(s) about $f(x)$ is (are) correct ?

(A) $f(x)$ has local minima at $x = 0$.

(B) $f(x)$ has local maxima at $x = 0$.

(C) Absolute maximum value of $f(x)$ is not defined.

(D) $f(x)$ is local maxima at $x = -3, x = 1$.

MMM159

12. A rectangular sheet of fixed perimeter with sides having their lengths in the ratio 8 : 15 is converted into an open rectangular box by folding after removing squares of equal area from all four corners. If the total area of removed squares is 100, the resulting box has maximum volume. The lengths of the sides of the rectangular sheet are

(A) 24 (B) 32 (C) 45 (D) 60

MMM160

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	C	C	C	B	D	B	D	D	A	C
Question	11	12								
Answer	A,C,D	A,C								

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Important Notes

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